Practice workbook · Turning sin and cos equations into tan

Rearranging an equation with sine and cosine of the same angle (possibly after expanding compound angles) into tan = k, then solving.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-25

1. Question 2 · part whole · 5 marks

2022 JAN · Paper 02

Worked-example retry · Attempt part whole only.

2 Solve the equation

$$\sin(2\theta - 20)° - \sqrt{3}\cos(2\theta - 20)° = 0 \quad \text{for } 0 \leqslant \theta \leqslant 180$$

(5)

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1. Question 2 · part whole · 5 marks

2022 JAN · Paper 02

Worked-example retry · Attempt part whole only.

Question 2 continued

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(Total for Question 2 is 5 marks)

1. Question 2 · part whole · 5 marks · Working and answer

Working and answer

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2. Question 10 · part b · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

10 (a) Solve the equation

$$\sin\left(x + \frac{\pi}{3}\right) = \frac{\sqrt{3}}{2} \qquad \text{for } 0 \leqslant x \leqslant 2\pi$$

Give your solutions in terms of π, where appropriate.

(3)

(b) Solve the equation

$$3\sin\theta + 5\cos\theta = 0 \qquad \text{for } -360° \leqslant \theta \leqslant 360°$$

Give your solutions to the nearest degree.

(3)

(c) Solve the equation

$$1 + \sin 2y = 2\cos^2 2y \qquad \text{for } -180° \leqslant y \leqslant 0°$$

(5)

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2. Question 10 · part b · 3 marks · Working and answer

Working and answer

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3. Question 10 · part a · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

10

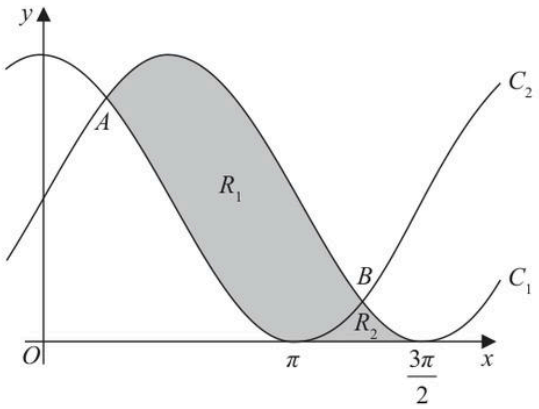


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows part of the curve C_1 with equation $y = \sin x + 1$ and part of the curve C_2 with equation $y = \cos x + 1$

As shown in Figure 2, C_1 and C_2 intersect at the point A and at the point B

(a) Find the exact value of the x coordinate of A and the exact value of the x coordinate of B **(3)**

The shaded finite region R_1 shown in Figure 2 is bounded by C_1 and C_2

The shaded finite region R_2 shown in Figure 2 is bounded by the x-axis, C_1 and C_2

(b) Use calculus to find the ratio

$$\text{area of } R_1 : \text{area of } R_2$$

Give your answer in the form $a : \left(\frac{\pi\sqrt{2}}{b} - c\right)$ where a, b and c are integers.

(9)

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3. Question 10 · part a · 3 marks · Working and answer

Working and answer

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4. Question 12 · part b · 2 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

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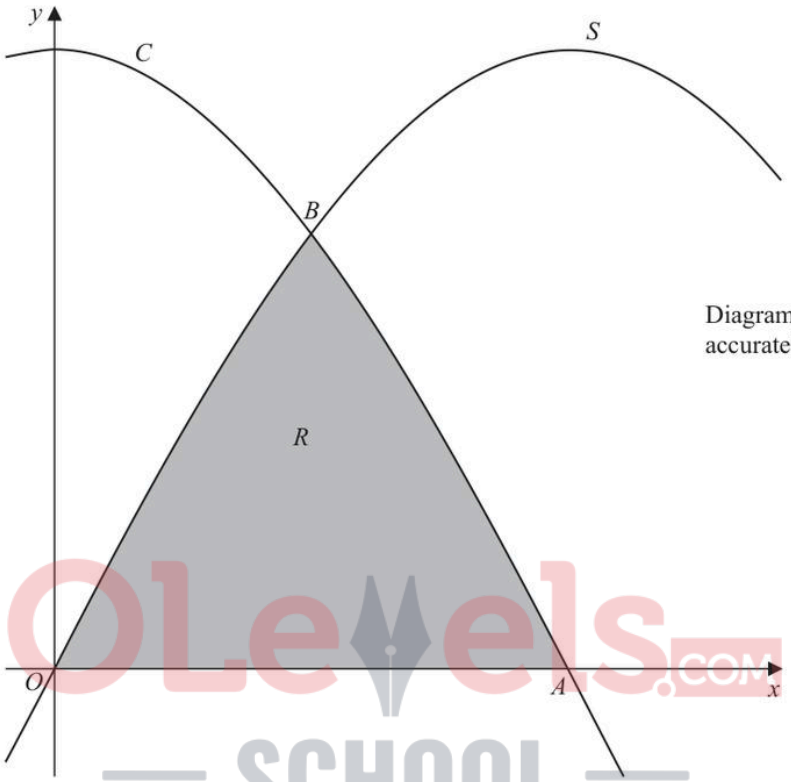


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Figure 2

The region R, shown shaded in Figure 2, is bounded by the x-axis, the curve S with equation $y = 2\sin x$ and the curve C with equation $y = 2\cos x$.
As shown in Figure 2, C crosses the x-axis at the point A.

(a) Write down the x coordinate of A.

(1)

As shown in Figure 2, C and S intersect at the point B.

(b) Find the x coordinate of B.

(2)

(c) Using calculus, find the area of the shaded region R.
Give your answer in the form $a - \sqrt{b}$ where a and b are integers.

(4)

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4. Question 12 · part b · 2 marks · Working and answer

Working and answer

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5. Question 9 · part c · 4 marks

2023 JUN · Paper 01

Further practice · Attempt part c only.

9 (a) Using the formulae on page 2, show that

(i) $\cos^2 A = \dfrac{\cos 2A + 1}{2}$

(ii) $\sin^2 A = \dfrac{1 - \cos 2A}{2}$

(4)

(b) Show that

$$(2\sin x - \cos x)(\sin x - 3\cos x) = \frac{1}{2}(\cos 2x - 7\sin 2x + 5)$$

(5)

$$y = (2\sin x - \cos x)(\sin x - 3\cos x)$$

(c) Solve, for $0° \leqslant x \leqslant 180°$ the equation, $\dfrac{dy}{dx} = 0$

Give your answers to the nearest whole number.

(4)

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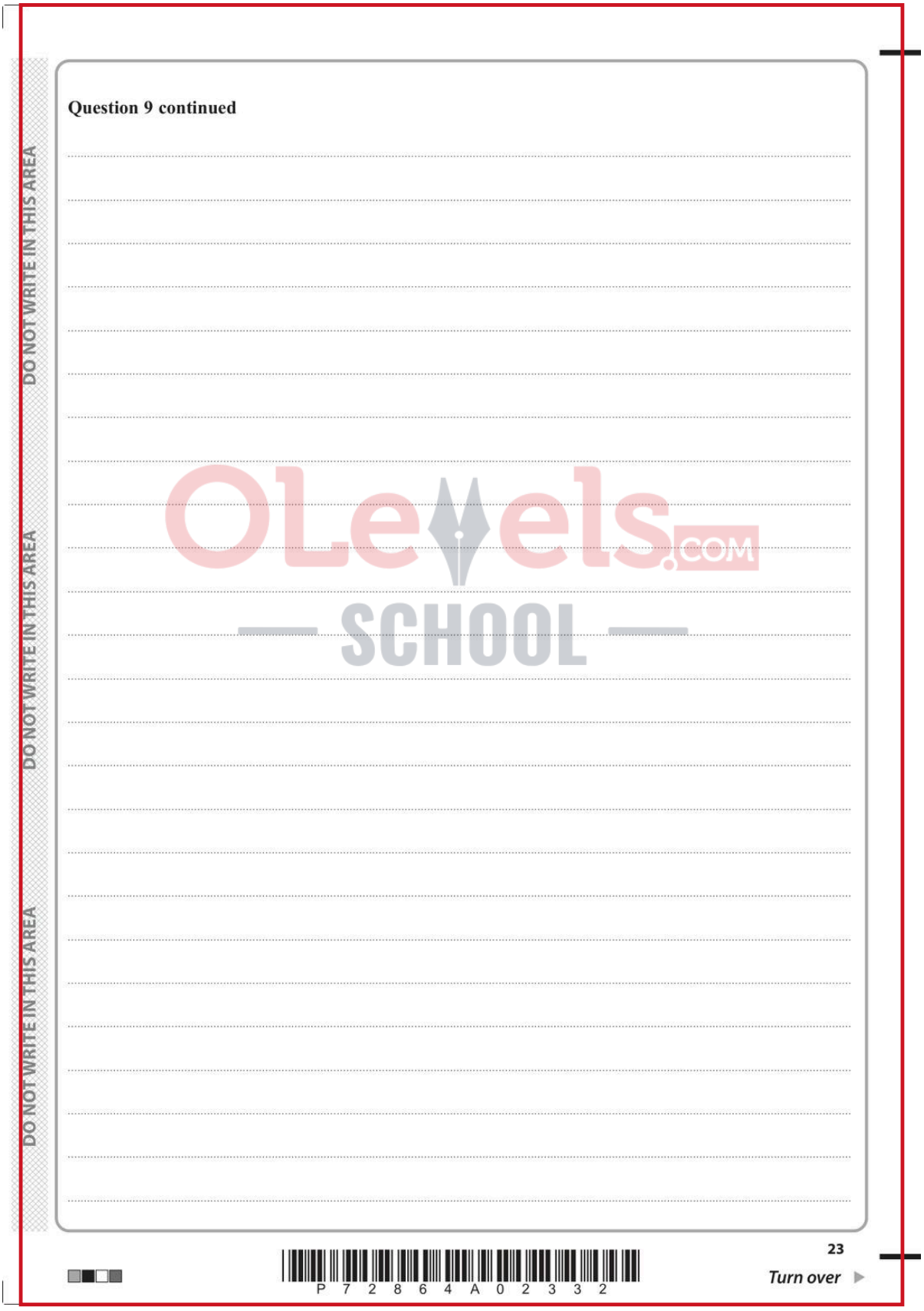
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5. Question 9 · part c · 4 marks

2023 JUN · Paper 01

Further practice · Attempt part c only.



5. Question 9 · part c · 4 marks

2023 JUN · Paper 01

Further practice · Attempt part c only.

Question 9 continued

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5. Question 9 · part c · 4 marks

2023 JUN · Paper 01

Further practice · Attempt part c only.

Question 9 continued

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(Total for Question 9 is 13 marks)

5. Question 9 · part c · 4 marks · Working and answer

Working and answer

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6. Question 11 · part b · 5 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

11 $f(\theta) = (2\cos\theta - \sin\theta)(2\sin\theta + \cos\theta)$

(a) Show that $f(\theta) = \frac{3}{2}\sin 2\theta + 2\cos 2\theta$

(3)

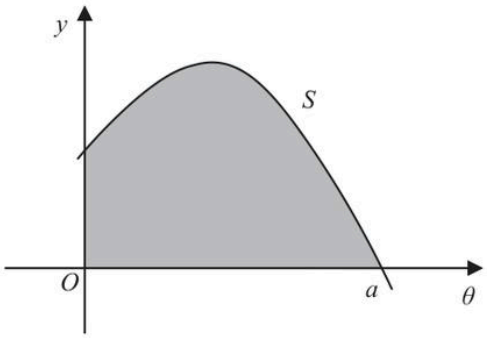


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Figure 4

Figure 4 shows part of the curve S with equation $y = f(\theta) + 2$

Given that S intersects with the θ-axis at the point with coordinates $(a, 0)$

(b) using $\sin^2\theta + \cos^2\theta = 1$, or otherwise, show that $a = \frac{\pi}{2}$

(5)

(c) Using algebraic integration, find the exact area bounded by S, the positive θ-axis and the positive y-axis shown shaded in Figure 4

(3)

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6. Question 11 · part b · 5 marks · Working and answer

Working and answer

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7. Question 8 · part b.ii · 4 marks

2024 NOV · Paper 01

Further practice · Attempt part b.ii only.

8 (i) (a) Using a formula given on page 2, show that

$$\tan 2A = \frac{2\tan A}{1-\tan^2 A}$$

(2)

(b) Hence, solve the equation

$$\tan A° - \tan 2A° = 0 \qquad \text{for } 0 \leqslant A \leqslant 180$$

(5)

(ii) Using a formula given on page 2, solve, giving your solutions as exact values

$$\cos\left(x - \frac{\pi}{6}\right) = \sin x \qquad \text{for } -\pi \leqslant x \leqslant 2\pi$$

(4)

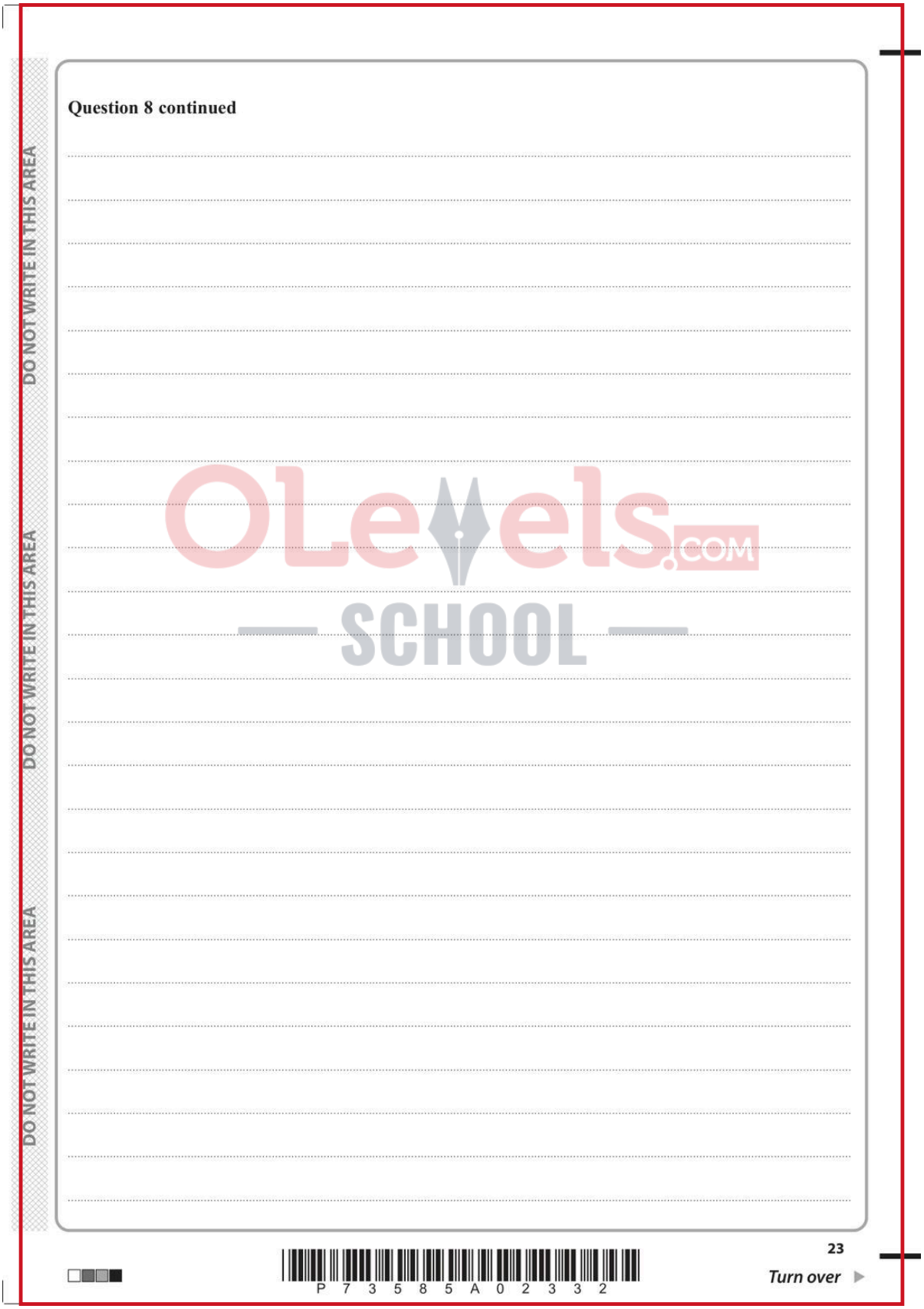
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7. Question 8 · part b.ii · 4 marks

2024 NOV · Paper 01

Further practice · Attempt part b.ii only.



Question 8 continued

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Turn over

7. Question 8 · part b.ii · 4 marks

2024 NOV · Paper 01

Further practice · Attempt part b.ii only.

Question 8 continued

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7. Question 8 · part b.ii · 4 marks

2024 NOV · Paper 01

Further practice · Attempt part b.ii only.

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Question 8 continued

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(Total for Question 8 is 11 marks)

7. Question 8 · part b.ii · 4 marks · Working and answer

Working and answer

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8. Question 7 · part b.ii · 4 marks

2022 JUN · Paper 02

Further practice · Attempt part b.ii only.

7 (i) (a) Using a formula from page 2, show that

$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

(2)

Given that $\tan 2\alpha = 1$

(b) show that $\tan\alpha = a \pm \sqrt{b}$ where a and b are integers whose values need to be found.

(3)

(ii) (a) Using formulae from page 2, show that $\cos(x-30)^\circ = \sin(x+30)^\circ$ can be written as $\tan x^\circ = 1$

(4)

(b) Hence, or otherwise, solve

$$\cos(2y-30)^\circ = \sin(2y+30)^\circ \qquad \text{for } -90 < y \leqslant 90$$

(2)

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8. Question 7 · part b.ii · 4 marks · Working and answer

Working and answer

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9. Question 11 · part a · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part a only.

11 (a) Express the equation

$$3\sin(A - B) = \sin(A + B)$$

in the form $\tan A = k \tan B$, giving the value of the integer k.

(4)

(b) Given that $\theta \neq \frac{(2n+1)\pi}{2}$ where $n \in \mathbb{Z}$,

show that $\frac{\cos^4\theta - \sin^4\theta}{\cos^2\theta} = 1 - \tan^2\theta$

(3)

(c) Using the exact values of $\sin x°$, $\cos x°$ and $\tan x°$ for $x = 30, 45, 60$

show that

(i) $\cos 15° = \frac{\sqrt{6} + \sqrt{2}}{4}$

(2)

(ii) $\tan 255° = \frac{3 + \sqrt{3}}{3 - \sqrt{3}}$

(4)

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9. Question 11 · part a · 4 marks · Working and answer

Working and answer

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